Entropy and Normalization Certificates for Low-Temperature SK Proposals

Exact Gaussian Target Costs and Typical-Fixed-Start Boundaries

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Abstract

We examine the September 24, 2026 typical-start upper-bound manuscript for zero-field Gaussian Sherrington–Kirkpatrick heat-bath dynamics at fixed inverse temperature greater than one. The initial state is one realized Gibbs draw, not a stationary average inside total variation. An exact Gaussian integration-by-parts identity replaces the source's square-root smoothing cost in its averaged target/base entropy comparison by a linear cost. Combining this identity with the actual strongly log-concave conditional disorder law gives an explicit coverage budget and a one-dimensional optimization certificate. A separate likelihood perturbation bound controls terminal field approximations on the source's enlarged proposal space. Positive normalized scalar panels retain the covariance term and provide local moment and tail error bounds. These deductions do not replace the source's Parisi, Gaussian-query, connection, or balanced-flow inputs, and do not improve its final mixing exponent here. Contribution assessment: C3/5, a sharper source-specific reverse-entropy budget with law-preserving approximation safeguards, not an independent low-temperature mixing proof.

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1 The Fixed-State Question

Fix $n \geq 2$ and a finite $\beta > 1$. The off-diagonal J_{ij}, $i < j$, are independent $N(0,1)$, and

$$H_J(\sigma) = n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j, \qquad \pi_J(\sigma) = Z_J^{-1} e^{\beta H_J(\sigma)}, \quad \sigma \in \{-1,1\}^n.$$

There is no external field. Each site has a rate-one clock; a ring resamples that spin from its conditional Gibbs law, including the possibility of retaining its old value. Thus the total update-attempt rate is n, not one.

For a fixed disorder and a fixed state, write

$$T_J(\sigma) = \inf\{t : \|P_t^J(\sigma,\cdot) - \pi_J\|_{\mathrm{TV}} \leq 1/4\}.$$

The assigned source [1] claims

$$\Pr_{J,\ \sigma_0\sim\pi_J} \{T_J(\sigma_0) \leq \exp(n^{1-1/40000000})\} \longrightarrow 1. \tag{1}$$

The one Gibbs draw is held fixed before taking the distance. In contrast, $\sum_\sigma \pi_J(\sigma)P_t^J(\sigma,\cdot) = \pi_J$ by stationarity, so moving the Gibbs average inside the norm would erase the question. A worst-start bound is a different quantifier and is not asserted here.

Current primary context. Sellke's primary Theorem 1.1 [3] gives exponentially slow worst-start mixing at sufficiently low temperature and explicitly distinguishes random initialization. We use that distinction, not its numerical temperature normalization. The assigned source invokes Lopatto's version 3 of the full interval-support result. The current version 4, dated October 5, 2026, still states support $[0, q_\beta]$, a smooth density below the top point and a top atom [2]. Its Theorem 1.1 and introductory proof outline were read, not its full proof. The revised external proof is not silently substituted for the pinned source's numbered dependencies.

2 Source Mechanisms and Exact Laws

Complete the Gaussian matrix with independent diagonal entries of variance $2/n$. Let G be its independent standard Gaussian drivers, γ their law, and put

$$\widehat{H}_G(x) = \tfrac{1}{2}x^{\mathsf{T}}W(G)x, \qquad F(G) = \log\sum_x e^{\beta\widehat{H}_G(x)}, \qquad P(dG,x) = \gamma(dG)\pi_G(x).$$

The diagonal contributes a linear function of G independent of x; it cancels from π_G, the dynamics, and the Hessian of F.

Source module	Mechanism retained in this report
Scalar	Full interval support, scalar stationarity, pressure upper bound, and continuation of a bounded magnetization martingale.
Proposal	Orthogonal adaptive matrix queries, fresh Gaussian completions and bridges, an independent noise channel, and terminal spins.
Reverse	Conditional strong log-concavity, Gaussian smoothing, and the conditional mixture identity for target/base coverage.
Connections	Independent fine endpoints conditional on a common base, coarse intermediate proposals, rotation, and boundary avoidance.
Mixing	Prune a small stationary mass, prove retained-set Poincare control, then dissipate entropy in the full chain.

All eleven cached public text files, including these complete proofs, were freshly read. This is a reading scope, not a certification of all cited results or of the global conclusion.

2.1 The scalar balance and its proposal

In variance time $v = \beta^2 s$, the source uses

$$\Phi_v = -\tfrac{1}{2}(\Phi_{yy} + \alpha_v \Phi_y^2), \qquad dY_v = \alpha_v m(v, Y_v)\, dv + dZ_v, \quad m = \Phi_y, \quad a = \Phi_{yy}.$$

The imported support is $[0, q_*]$, with $v_* = \beta^2 q_*$. On $[0, v_*]$, $dm_v = a_v\, dZ_v$ and $\mathbb{E}m_v^2 = v/\beta^2$. Beyond v_*, $m(v, y) = \tanh y$ and $a(v, y) = \operatorname{sech}^2 y$. Set $p(v) = \mathbb{E}a_v$, $r(v) = \mathbb{E}a_v^2$, and $\rho(v) = \beta\sqrt{r(v)}$. The symbol $\rho(v)$ here is the scalar noise coefficient; the constant correlation parameter used later will be denoted r_0.

The source proves $0 < a \le 1 - m^2$, $r = \beta^{-2}$ before v_*, and on the continuation

$$p' = -r, \quad r' = 4r - 6\mathbb{E}a_v^3, \quad r/p \ge 1/2, \quad \mathbb{E}a_v^3/r \ge 3/4.$$

In particular $p(v) \le C_\beta e^{-v/2}$, $-2 \le (\log r)' \le 0$, and $0 \le \rho \le 1$. The displayed cubic ratio in fact gives $r' \le -r/2$; this elementary strengthening is a secondary consequence, not the principal contribution. For

$$e(V) = \int_0^V \sqrt{r}\, p\, \rho\, dv, \qquad j(V) = \tfrac{1}{2}\int_0^V \rho^2 \mathbb{E}\frac{a_v^2}{1 - m_v^2}\, dv,$$

the exact scalar identity is

$$\log 2 - j(V) + \beta e(V) = \mathcal{P} - \frac{\beta^2}{4} p(V)^2. \tag{2}$$

Its proof uses the binary entropy Ito identity up to v_*; above v_* both sides have derivative $\beta^2 rp/2$. The ratio near pinning is legitimate because $a \le 1 - m^2$. It is not legitimate for arbitrary independent approximations of its numerator and denominator.

At mesh $h = n^{-1/16}$ and horizon $V \sim l \log n$, the proposal queries normalized orthogonalized scalar increments. Conditional on earlier columns, the remaining matrix block is GOE; Gaussian completion fills its missing directions. Together with bridges this gives a genuine Brownian stream B. An independent D enters through $dZ = \rho\, dB + \sqrt{1 - \rho^2}\, dD$. The final spins are independent given the paths, with means $m(V, Y_{V,i})$. The resulting joint law Q_l has Gaussian disorder marginal and uniform spin marginal, but its spin and disorder are not independent.

The source's enlarged-space linking density is $\prod_i (1 + x_i m_{V,i})$. Splitting its stochastic exponential into the B and D channels and checking the enlarged filtration gives an auxiliary reference with

independent Gaussian disorder and uniform spin. Multiplying by the Gibbs density gives a reference with marginal P. The resulting log likelihood combines $nj(V)$, the proposal energy and $F(G)$; (2) cancels the leading terms. The source establishes separately

$$D_l := \mathrm{KL}(Q_l \| P) \le 2n^{1-l/2} \tag{3}$$

and a good-output subprobability dominated by $e^{n^{1-l/2}}P$, with exceptional probability at most $e^{-n^{1-10l}}$. Forward entropy alone cannot replace this likelihood assertion.

3 An Exact Gaussian Target/Base Cost

We first prove a finite-dimensional identity requiring neither the Parisi theorem nor a mixing assertion. Let $\mathcal{X}$ be finite and

$$\pi_g(x) = e^{b_x \cdot g}/Z(g), \qquad F(g) = \log Z(g), \qquad g \in \mathbb{R}^N, \quad b_x \in \mathbb{R}^N.$$

Constant positive base weights can also be included in Z. All derivatives exist, $\nabla F = \pi_g b$, and $\nabla^2 F = \mathrm{Cov}_{\pi_g}(b)$. In particular gradients and Hessians are bounded for fixed finite $\mathcal{X}$.

Theorem 3.1 (Gaussian Bregman identity)**.** *Let G, Z be independent standard Gaussian vectors and $G^\circ = r_0 G + \sqrt{1-r_0^2}\, Z$, $0 \le r_0 < 1$. Then*

$$\mathbb{E}\,\mathrm{KL}(\pi_G \| \pi_{G^\circ}) = (1-r_0)\mathbb{E}\,\mathrm{tr}\,\nabla^2 F(G). \tag{4}$$

For the SK normalization above, define $R_{12} = n^{-1}\sum_i x_i^{(1)} x_i^{(2)}$, with replicas independent conditional on G. Then

$$\begin{aligned} C_n := \mathbb{E}\,\mathrm{tr}\,\nabla^2 F(G) &= \frac{\beta^2}{n}\sum_{i<j}\mathbb{E}[1-\pi_G(x_i x_j)^2] \\ &= \frac{\beta^2 n}{2}(1-\mathbb{E}\pi_G^{\otimes 2} R_{12}^2) \le \frac{\beta^2(n-1)}{2}. \end{aligned} \tag{5}$$

Consequently, with $\delta = 1-r_0^2$,

$$\mathbb{E}\,\mathrm{KL}(\pi_G \| \pi_{G^\circ}) = \frac{\delta}{1+\sqrt{1-\delta}} C_n \le \frac{\beta^2(n-1)}{2}\,\delta. \tag{6}$$

Proof. The Gibbs logarithmic ratio gives the exact Bregman formula

$$\mathrm{KL}(\pi_g \| \pi_h) = F(h) - F(g) - \nabla F(g)\cdot(h-g).$$

Since G° and G have identical Gaussian marginals, the expected F terms cancel. Conditional expectation of G° given G is $r_0 G$, so the remaining quantity is $(1-r_0)\mathbb{E}[G\cdot\nabla F(G)]$. Gaussian integration by parts makes this $(1-r_0)\mathbb{E}\,\mathrm{tr}\,\nabla^2 F(G)$. Bounded derivatives and the at-most-linear growth of F justify every expectation.

For an off-diagonal driver, $b_{x,ij} = \beta x_i x_j/\sqrt{n}$. For a diagonal driver the coefficient is independent of the spin, hence its covariance is zero. Taking the Hessian trace gives the first line of (5). Replica independence gives

$$\pi_G^{\otimes 2} R_{12}^2 = \frac{1}{n^2}\left(n + 2\sum_{i<j}\pi_G(x_i x_j)^2\right),$$

which gives its second line. Each summand in the first line lies in $[0,1]$. Finally $1-r_0 = \delta/(1+r_0) \le \delta$. □

This improves the source's averaged $O_\beta(n\sqrt{\delta})$ target/base bound to an explicit linear smoothing cost. It is an application of classical Gaussian integration by parts and exponential-family covariance, not a new general identity. The important source-specific distinction is the expectation *before* taking the supremum over spins. The same improvement cannot be used for a pointwise supremum of energy changes.

Example 3.2 (Two spins and a nonzero cost). For $n = 2$, write $u = \beta J_{12}/\sqrt{2}$. Ignoring the cancelled diagonal, $F = \log(4\cosh u)$, and

$$C_2 = \frac{\beta^2}{2}\mathbb{E}\operatorname{sech}^2(\beta J_{12}/\sqrt{2}).$$

Formula (4) is therefore an exact nonzero Gaussian expectation identity for this genuine SK system. No closed form or numerical evaluation of this integral is claimed. The pointwise covariance and replica identities can be tested with rational Gibbs weights; the Gaussian integration step is proved analytically, not inferred from finite samples.

4 Reverse Entropy and Simultaneous Coverage

Let Q be any joint law with disorder marginal γ, uniform spin marginal, and $D = \mathrm{KL}(Q\|P) < \infty$. The source Q_f is one example; an approximation must verify these same premises. Define $\mathcal{S}_\delta$ by replacing the disorder by $r_0 G + \sqrt{\delta}Z$, retaining the spin, where $r_0 = \sqrt{1-\delta}$ and $0 < \delta < 1$. Write $\nu(\cdot \mid G^\circ)$ for the conditional spin law of $\mathcal{S}_\delta Q$.

Lemma 4.1 (The actual conditional reference)**.** *For every spin x, $P(dG \mid x)$ is strongly log-concave with curvature at least one, and*

$$\mathrm{KL}(\mathcal{S}_\delta P\|\mathcal{S}_\delta Q) \le \frac{1-\delta}{\delta}D. \tag{7}$$

Proof. Coordinate-sign gauge symmetry of Gaussian SK makes P_X uniform. The conditional density is $2^n\gamma(G)\pi_G(x)$; its negative logarithm, up to a constant, is

$$U_x(G) = \tfrac{1}{2}\|G\|^2 - \beta\widehat{H}_G(x) + F(G), \qquad \nabla^2 U_x = I + \mathrm{Cov}_{\pi_G}(b) \succeq I.$$

This is not an assertion that π_G on the spin cube is strongly log-concave, nor that the conditional law is Gaussian.

The classical quadratic transport inequality for this reference gives $W_2^2(Q(\cdot \mid x), P(\cdot \mid x)) \le 2\,\mathrm{KL}(Q(\cdot \mid x)\|P(\cdot \mid x))$. To recall the source's finite-dimensional justification, transport the reference to the other density by an increasing triangular quantile map T on a box. Strong convexity in the entropy change-of-variables formula supplies $\|T(z) - z\|^2/2$. Integration by parts turns the remaining linear term into $\mathrm{div}(T - I)$; the residual diagonal Jacobian terms are $a - 1 - \log a \ge 0$. The normal displacement vanishes on box faces. Truncation, positive smooth approximation, tightness and lower semicontinuity remove the box. Here both conditional laws are bounded above by $2^n\gamma$, and the reference has Gaussian tails up to a linear exponential factor, so the entropy and moment approximations are justified for fixed n.

For any coupled pair $g, \widetilde{g}$, the divergence between the Gaussian smoothing kernels, in either orientation, is

$$\mathrm{KL}(N(r_0 g, \delta I)\|N(r_0\widetilde{g}, \delta I)) = \frac{r_0^2}{2\delta}\|g - \widetilde{g}\|^2.$$

Convexity under mixing bounds the divergence of the two smoothed conditional laws by this coupling cost. Average over the *common* uniform spin marginal and use the entropy chain rule. This proves (7), retaining the factor $1 - \delta$ that the source safely bounded by one. □

Theorem 4.2 (Sharpened target/base budget)**.** *Sample G, Z independently from γ and form G° as above. Under the preceding premises,*

$$\mathbb{E}\,\mathrm{KL}(\pi_G \| \nu(\cdot \mid G^\circ)) \le \mathcal{B}(r_0) := C_n(1-r_0) + D\frac{r_0^2}{1-r_0^2}. \tag{8}$$

For any $L > 0$ and $0 < a < 1$, outside an event of target/base probability at most $\mathcal{B}(r_0)/L$, simultaneously for every set A chosen after that pair is realized,

$$\pi_G(A) \ge a \quad \Longrightarrow \quad \nu(A \mid G^\circ) \ge \exp\!\left(-\frac{L + H_2(a)}{a}\right), \tag{9}$$

where $H_2(a) = -a\log a - (1-a)\log(1-a)$.

Proof. Let $\overline{\pi}(\cdot \mid G^\circ) = \mathbb{E}[\pi_G \mid G^\circ]$. Insert this conditional mixture into the logarithmic ratio:

$$\begin{aligned}\mathbb{E}\,\mathrm{KL}(\pi_G \| \nu_{G^\circ}) &= \mathbb{E}\,\mathrm{KL}(\pi_G \| \overline{\pi}_{G^\circ}) + \mathrm{KL}(\mathcal{S}_\delta P \| \mathcal{S}_\delta Q) \\ &\le \mathbb{E}\,\mathrm{KL}(\pi_G \| \pi_{G^\circ}) + \mathrm{KL}(\mathcal{S}_\delta P \| \mathcal{S}_\delta Q).\end{aligned}$$

The inequality is the minimizing property of the conditional mixture, itself obtained from the same identity for any base-measurable spin law. Apply Theorem 3.1 and Lemma 4.1. Markov's inequality gives a single event on which the conditional divergence is at most L.

On that event put $s = \pi_G(A)$, $t = \nu(A \mid G^\circ)$. Data processing to the partition $\{A, A^c\}$ gives

$$s\log(1/t) \le \mathrm{KL}(\pi_G \| \nu) + H_2(s).$$

The function $(L + H_2(s))/s$ is decreasing on $(0,1)$, since its derivative is $(\log(1-s) - L)/s^2 < 0$. For $s \ge a$ this proves (9); $s = 1$ follows by continuity. No union bound over sets is used. □

4.1 Optimizing only the entropy module

Any certified upper bounds $C \ge C_n$, $D_0 \ge D$ can be substituted in (8). For $C, D_0 > 0$, the function

$$B(r) = C(1-r) + D_0 r^2/(1-r^2), \qquad 0 \le r < 1,$$

has a unique minimum in $(0,1)$. Indeed

$$B'(r) = -C + \frac{2D_0 r}{(1-r^2)^2}, \qquad B''(r) = \frac{2D_0(1+3r^2)}{(1-r^2)^3} > 0. \tag{10}$$

The minimizer is the unique root $C(1-r^2)^2 = 2D_0 r$. Rational endpoint signs bracket it without claiming to solve a stochastic model. When $D_0 = 0$, the infimum is zero as $r \uparrow 1$; when $C = 0$, choose complete smoothing. Complete smoothing is a valid endpoint since it makes both spin kernels uniform.

A simpler explicit sufficient budget is

$$B \le C\delta + D_0/\delta. \tag{11}$$

If $0 < D_0 \le C$, choose $\delta = \sqrt{D_0/C}$ to get $2\sqrt{CD_0}$. This is a sufficient choice, not the exact optimizer of (8). The earlier square-root cost $C\sqrt{\delta} + D_0/\delta$ has a different small-error power, of order $C^{2/3}D_0^{1/3}$. For illustration, $C = 2$, $D_0 = 2 \cdot 10^{-8}$, $\delta = 10^{-4}$ give the new sufficient bound $4 \cdot 10^{-4}$, whereas the square-root expression at the same smoothing level is 0.0202. These are exact budget values, not measured proposal errors or running times.

Corollary 4.3 (Coverage exponents with source parameters)**.** *Retain the source's $f = 1/100$, $D_f \le 2n^{1-f/2}$, and $\delta = n^{-f/4}$. Then*

$$\mathbb{E}\,\mathrm{KL}(\pi_G \| \nu_{G^\circ}) \le (\beta^2/2 + 2) n^{1-f/4}.$$

For fixed $0 < d < \kappa < f/4$, with failure probability at most $(\beta^2/2+2)n^{-(f/4-\kappa)}$, every set of Gibbs mass at least n^{-d} has proposal mass at least

$$\exp\{-n^{1-\kappa+d} - H_2(n^{-d})n^d\}.$$

In particular any $\gamma_0 < \kappa - d$ gives the weaker lower bound $\exp(-n^{1-\gamma_0})$ for sufficiently large n.

Proof. Apply (6) and (7), then Theorem 4.2 with $L = n^{1-\kappa}$. The elementary inequality $-(1-a)\log(1-a) \le a$ shows $H_2(n^{-d})n^d \le d\log n + 1$, which is negligible at the stated power. □

The source's fixed $d = f/100$, $\kappa = 4d$ remain valid. The expectation improves from its $n^{1-f/8}$ scale to $n^{1-f/4}$ at the same δ. This concerns only coverage. In particular the connections proof still requires a high-probability energy supremum of order $n\sqrt{\delta}$; an averaged identity does not replace that supremum.

5 A Law-Preserving Approximation Budget

The previous section accepts any proposal with the two correct marginals and a forward-entropy certificate. We now quantify one restricted approximation which keeps the original query experiment rather than replacing it by an unrelated Gaussian law.

Let Q be the source's linked enlarged experiment, including all query and auxiliary paths and the final spin. Let R be its enlarged reference whose (G, x) marginal is P. Write

$$L = \log(d\mathsf{Q}/d\mathsf{R}), \quad D_{\mathrm{ext}} = \mathbb{E}_{\mathsf{Q}} L, \quad \sigma_L^2 = \mathrm{Var}_{\mathsf{Q}}\, L < \infty.$$

The source proves its forward comparison by bounding this enlarged log likelihood before marginalization. These quantities are properties of the *actual* experiment.

Theorem 5.1 (Terminal-field perturbation)**.** *Keep the entire path and disorder law unchanged. At a horizon $V > v_*$, replace only the conditional terminal spin means $\tanh Y_i$ by $\tanh(Y_i + e_i)$, where e_i is measurable from that path and $|e_i| \le \varepsilon_i$ almost surely, with deterministic finite ε_i. Let $\widehat{\mathsf{Q}}$ be this experiment and $\widehat{Q}$ its (G, x) marginal. Put $\omega = 2\sum_i \varepsilon_i$. Then*

$$\mathrm{KL}(\widehat{Q} \| P) \le D_{\mathrm{ext}} + 2\sinh(\omega/2)\sigma_L + \tfrac{1}{2}\sum_i \mathbb{E} e_i^2. \tag{12}$$

The disorder marginal stays γ. To use this bound in Theorem 4.2, the uniform spin marginal must additionally be verified, for example by coordinate-sign gauge equivariance of the full perturbed experiment.

Proof. Let $u = d\widehat{\mathsf{Q}}/d\mathsf{Q}$. For a fixed path, the terminal laws are products of signs in fields Y_i and $Y_i + e_i$, respectively. Their log ratio is

$$\log u = \sum_i \{x_i e_i - \log\cosh(Y_i + e_i) + \log\cosh Y_i\}.$$

Since $|(\log\cosh)'| \le 1$, $e^{-\omega} \le u \le e^{\omega}$. The path law is unchanged, so $\mathbb{E}_{\mathsf{Q}} u = 1$. For a random variable in $[m, M]$ with mean one, $\mathbb{E}(u-1)^2 \le (M-1)(1-m)$, obtained by integrating $(u-m)(M-u) \ge 0$.

Here this is $4\sinh^2(\omega/2)$. Also the Bernoulli exponential-family divergence for a field shift e_i is a Bregman remainder of $\log\cosh$, whose second derivative is at most one. Thus

$$\mathrm{KL}(\widehat{\mathsf{Q}}\|\mathsf{Q}) \le \tfrac{1}{2}\sum_i \mathbb{E}e_i^2.$$

The expectation can be taken under either enlarged law on paths. Finally the exact entropy decomposition is

$$\mathrm{KL}(\widehat{\mathsf{Q}}\|\mathsf{R}) = D_{\mathrm{ext}} + \mathbb{E}_{\mathsf{Q}}[(u-1)(L-D_{\mathrm{ext}})] + \mathrm{KL}(\widehat{\mathsf{Q}}\|\mathsf{Q}).$$

Cauchy–Schwarz and marginalization prove the bound. Because only the terminal conditional draw changed, the disorder marginal did not change. □

The use of a field error is deliberate. A small absolute error in a near-pinned magnetization can set an originally positive spin probability to zero; then reverse relative entropy can be infinite. A bounded field shift preserves both signs and has a uniform normalized likelihood bound. On the other hand, ω adds over all coordinates and the sinh cost can be large. The theorem does not assert that an approximate query path has a small terminal field error, nor that changing the scalar coefficients leaves the adaptive query law unchanged. Those changes require a separate path-law analysis.

Gauge equivariance is a concrete sufficient marginal condition: for every $\eta \in \{-1,1\}^n$, transform off-diagonal drivers by $G_{ij} \mapsto \eta_i\eta_j G_{ij}$ and the output spin by $x_i \mapsto \eta_i x_i$, together with the corresponding transformation of auxiliary inputs. If the complete perturbed experiment is equivariant, Gaussian invariance makes its spin law invariant under this transitive action, hence uniform. Symmetry of a single scalar formula is not by itself a proof of equivariance of an adaptive implementation.

6 Positive Scalar Panels with Normalization

This section supplies a local evaluation certificate, not a replacement of the source's adaptive Gaussian-query law. For one scalar Parisi step take $0 \le \lambda \le 1$, $h > 0$, and a smooth real profile f satisfying

$$|f'| \le 1, \qquad 0 < f'' \le 1 - (f')^2.$$

For $U \sim N(0,1)$, the exact tilted law is

$$d\mu_{\lambda,y}(u) = \frac{e^{\lambda(f(y+\sqrt{h}\,u)-f(y))}}{I_0(y)}\, d\gamma_1(u).$$

It is not the unweighted Gaussian law. Put

$$A_0 = e^{\lambda(f(y+\sqrt{h}\,u)-f(y))}, \quad A_1 = A_0 f', \quad A_2 = A_0 f'', \quad A_3 = A_0 (f')^2, \qquad I_j = \int A_j\, d\gamma_1,$$

where the profile derivatives are evaluated at $y + \sqrt{h}\,u$. Differentiating the normalized integral gives

$$(Tf)' = \frac{I_1}{I_0}, \qquad (Tf)'' = \frac{I_2}{I_0} + \lambda\left(\frac{I_3}{I_0} - \left(\frac{I_1}{I_0}\right)^2\right). \tag{13}$$

At $\lambda = 0$, $Tf = \mathbb{E}f(y + \sqrt{h}\,U)$; otherwise $Tf = f(y) + \lambda^{-1}\log I_0$. The covariance term is precisely the normalization derivative that must not be frozen. The original Ram Urysohn proof [5] derives this quotient/covariance rule, including the second derivative, on a neighborhood with positive denominator.

Proposition 6.1 (Certified scalar ratios)**.** *Suppose a positive quadrature with nodes u_i and weights $w_i > 0$, independent of y in the differentiation neighborhood, gives $\widehat{I}_j = \sum_i w_i A_j(u_i)$. Assume certified errors $|I_j - \widehat{I}_j| \le \epsilon_j$. Let*

$$d_* = \max(1, \widehat{I}_0 - \epsilon_0), \qquad \widehat{\mu}_j = \widehat{I}_j/\widehat{I}_0, \qquad \eta_j = \frac{\epsilon_j + |\widehat{\mu}_j|\epsilon_0}{d_*}.$$

Then

$$\begin{aligned} |(Tf)' - \widehat{m}| &\le \eta_1, \\ |(Tf)'' - \widehat{a}| &\le \eta_2 + \lambda(\eta_3 + 2\eta_1), \end{aligned} \tag{14}$$

where $\widehat{m} = \widehat{\mu}_1$ and $\widehat{a} = \widehat{\mu}_2 + \lambda(\widehat{\mu}_3 - \widehat{\mu}_1^2)$. Moreover the positive finite model preserves exactly

$$|\widehat{m}| < 1, \qquad 0 < \widehat{a} \le 1 - \widehat{m}^2 - (1-\lambda)\operatorname{Var}_{\widehat{\mu}}(f') \le 1 - \widehat{m}^2. \tag{15}$$

Here $\widehat{\mu}$ is the normalized tilted discrete law, not the original quadrature weights.

Proof. Convexity of f gives $\mathbb{E}[f(y + \sqrt{h}\,U) - f(y)] \ge 0$. Jensen therefore gives $I_0 \ge 1$, also at $\lambda = 0$. Thus $I_0 \ge d_*$, while $\widehat{I}_0 > 0$. The exact quotient difference is

$$\frac{I_j}{I_0} - \widehat{\mu}_j = \frac{(I_j - \widehat{I}_j) - \widehat{\mu}_j(I_0 - \widehat{I}_0)}{I_0},$$

which proves the ratio bound. Both exact and finite normalized means lie in $[-1, 1]$, so their squares differ by at most $2\eta_1$. This proves (14). Finally average $f'' \le 1 - (f')^2$ under the positive finite law and add $\lambda \operatorname{Var}(f')$. Strict positivity and the strict mean range follow from the hypotheses at the finitely many nodes. □

In particular, the finite model's normalized sign integrand $\widehat{\xi} = x\widehat{a}/(1 + x\widehat{m})$ satisfies $|\widehat{\xi}| \le 1 - x\widehat{m} \le 2$. This is a useful sign-before-division safeguard near pinning. It does not prove that the finite model has the same diffusion or entropy–energy identity as the source.

6.1 Tail and moment errors under the actual tilted law

For $R \ge \lambda\sqrt{h}$, the Lipschitz bound on f gives, for every $j = 0, 1, 2, 3$,

$$\int_{|u|>R} |A_j(u)|\, d\gamma_1(u) \le 2e^{\lambda^2 h/2}\overline{\Phi}(R - \lambda\sqrt{h}) \le e^{\lambda^2 h/2 - (R-\lambda\sqrt{h})^2/2}. \tag{16}$$

Here $\overline{\Phi}$ is the standard Gaussian upper tail. To prove this, use $|A_j| \le e^{\lambda\sqrt{h}|u|}$, complete the square separately on each half-line, and use $\overline{\Phi}(t) \le \frac{1}{2}e^{-t^2/2}$ for $t \ge 0$. For the last inequality, the derivative of $e^{t^2/2}\overline{\Phi}(t)$ is nonpositive by the elementary Gaussian Mills bound, and its value at zero is $1/2$.

Partition $[-R, R]$ into panels I with centers c_I. If a positive rule is exact on polynomials of degree at most N for the Gaussian measure restricted to each panel, and $|A_j^{(N+1)}| \le M_{j,I}$ on that panel, Taylor's integral remainder gives the additional error

$$\sum_I \frac{M_{j,I}}{(N+1)!} \left\{ \int_I |u - c_I|^{N+1}\, d\gamma_1(u) + \sum_{u_i \in I} w_i |u_i - c_I|^{N+1} \right\}. \tag{17}$$

The two Taylor polynomials cancel by exactness; their two remainders give the displayed sum. Certified moment errors or node-value errors must be added if exactness is approximate. This is the

finite local-moment argument of the original chapter [4], with a separate noncompact tail. Neither a finite degree nor agreement of two quadratures certifies the derivative majorants $M_{j,I}$. Included subchapters of that original file were not read.

There are two different convexity statements in play. The disorder law $P(dG \mid x)$ has curvature at least one. The scalar tilted u-law instead has potential curvature

$$1 - \lambda h f''(y + \sqrt{h}\, u) \geq 1 - \lambda h.$$

It is strongly log-concave under this estimate only when $\lambda h < 1$. Its covariance bounds cannot be justified by borrowing the dimension-independent disorder curvature. Likewise, positive quadrature preserves (15) for the evaluated profile, but not arbitrary changes to that profile. A profile computed by a nonlinear residual solve needs its own neighborhood, invariant-ball and inverse/contractivity certificate. The original TR6 theorem gives error at most residual divided by $1 - q$ when its full map has $q < 1$; no such contraction is asserted here for the complete Parisi or proposal construction.

7 What Reaches Mixing, and What Does Not

The source uses $f = 1/100$, $d = f/100$, $b = d/100$. Fine horizon f supplies endpoint coverage; coarse horizon b supplies sufficiently rare failures along the rotated sequence. A large set A can be selected after the target and base are realized because (9) holds simultaneously. Fine endpoints are conditionally independent given the common base. Intermediate proposals are correlated, and their good-output subprobabilities are not renormalized.

The connection argument still needs $\sup_x |\widehat{H}_G(x) - \widehat{H}_{G(\theta)}(x)| = O(n\sqrt{\delta})$ outside a small exceptional event. This pays a Gibbs likelihood ratio on a boundary selected from the target disorder. The exact averaged cost (4) cannot be substituted into this pathwise use. It also does not replace Gram concentration, adaptive orthogonalization, rare coarse failures, or the small-Hamming-step construction. Accordingly we retain the original balanced-flow threshold $B_{\min} = \exp(-n^{1-b/20})$.

For completeness, the final finite-chain implication has the following precise form. A reversible chain with outgoing rate at most R, and flow greater than B across every cut with both masses at least $\varepsilon < 1/3$, satisfies

$$\sum_x \pi(x) \|P_T(x, \cdot) - \pi\|_{\mathrm{TV}} \leq \frac{\sqrt{2RH(\pi)}}{B\sqrt{T}} + 2\varepsilon. \tag{18}$$

The source proves this by pruning nonexpanding sets until a set of mass greater than $1 - \varepsilon$ remains. Its unnormalized Poincare constant is at most $2R/B^2$. For the full-chain densities $f_t^x = P_t(x, \cdot)/\pi$, entropy dissipation gives

$$\int_0^T \sum_x \pi(x) \mathfrak{D}(\sqrt{f_t^x})\, dt \leq H(\pi)/4.$$

At one time, retained-set Cauchy–Schwarz controls the fixed-state distances; the complement has averaged mass at most 2ε by stationarity. Contraction moves the estimate to the prescribed time. No trajectory is assumed to stay in the retained set.

With $R = n$, $H(\pi) \leq n \log 2$, $\varepsilon = n^{-d}$, and the original $B_{\min}$, the source takes $T = \exp(n^{1-b/40})$. The resulting small *average of distances* and Markov's inequality give its typical fixed-start statement. Our coverage improvement leaves these final choices unchanged. An optimized coverage exponent alone is not an optimized mixing exponent.

8 Advantages, Limitations, and Further Work

The principal gain is a linear, explicit averaged smoothing cost with the actual overlap-dependent constant C_n. It improves the entropy budget without changing the proposal. The exact one-dimensional objective can be bracketed by rational arithmetic once C, D_0 are certified. The terminal-field bound and positive scalar panels expose what a finite approximation must preserve: normalization, positivity, the actual reference law, and both marginals.

These benefits have costs. The source's finite-volume proposal and Parisi inputs remain substantive. The variance σ_L^2 in (12) must be bounded for the actual enlarged likelihood; a residual for a scalar function alone does not provide it. Coordinate field errors accumulate, and the displayed bound may be conservative. The normalized scalar panel rule certifies local evaluation, not a complete approximation to adaptive matrix queries. All thresholds may depend on fixed β; no uniform low-temperature limit $\beta \to \infty$ is claimed. Gaussian integration by parts does not give the same exact identity for arbitrary non-Gaussian disorder.

Useful next steps are a certified overlap bound improving C_n, a variance-sensitive rather than uniform likelihood perturbation bound, and a complete error analysis for approximated query paths. Any reoptimization of the final mixing exponent must check the separate boundary, coarse failure, grid and Hamming-distance inequalities. We make no measured speedup or historical novelty claim for Gaussian integration by parts, transport, normalized covariance calculus, or Taylor remainders.

9 Read Scope and Assessment

All 1,752 lines in the eleven cached public text files were freshly displayed in bounded untruncated chunks, including the full scalar, proposal, reverse, connections and mixing proofs. Pre-read and post-read SHA-256 endpoints agree. The figure's TeX and caption were read; a separate rendering of the source figure was not inspected. Reading all source text is not a verification of every external theorem.

The original local-moment chapter was read at lines 1–446; its included files were not read. Original TR6 was read at 329–425, 648–750 and 2104–2142. Lines 648–649 close a previous formula; only the complete statements beginning thereafter support the contraction comparison. The rest of that 2,250-line file is explicitly unread. The later core16 file was hashed as a locator but not read or used as original proof evidence. Thus 685 original private lines, not an entire corpus, were freshly inspected. The audit ledger records every interval and unread complement. Private material remained local; web queries used only public titles and URLs.

Owned exact checks exercise pointwise Gibbs covariance and replica identities, normalized finite-law derivative identities, budget optimization signs, and likelihood-ratio inequalities. They do not discretize the Parisi minimizer, replay a foreign checker, or establish the Gaussian averaged identity by numerical sampling. Counts of parameter cases and assertions are separate in the frozen receipt. Parent checks are excluded.

Applicability is L3. Contribution C3 is assigned to the source-specific reverse-entropy improvement and its controlled approximation interface, not to generic finite-rank algebra. No source mathematical correction is asserted: its square-root bound and final theorem are not contradicted. The exact model and source premises are retained, and no independent proof of (1) is claimed.

References

[1] OpenAI, *A typical-start upper bound for low-temperature SK Glauber dynamics*, September 24, 2026. Public manuscript directory. Comparison uses the captured unchanged source, not a live-branch assertion.

[2] P. Lopatto, *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*, arXiv:2607.11756v4, October 5, 2026. Current primary metadata, Theorems 1.1–1.2 and introductory outline read; remaining proof not audited. The assigned source cites v3, July 16, 2026.

[3] M. Sellke, *Exponentially Slow Mixing of the Low Temperature SK Model*, arXiv:2511.22621v2, December 9, 2025. Primary metadata and returned scientific HTML through the proof and remark read. External gapped-state and restricted-matrix inputs not audited here.

[4] M. Subbarao, original confidential RamCalBooks, volume 1 rev2, `ch03-local-moments-rev2.tex`, lines 1–446. Local Taylor remainder, normalization and positivity limitations; included subchapters not read.

[5] M. Subbarao, original confidential `TR6-lin-non-lin.tex`, lines 329–425, 648–750, 2104–2142. Full variations, normalized quotient derivatives and conditional local residual control; remaining text unread.